

Global unprojected Cartan coefficient-ledger no-go and rational forest boundary

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

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1 Purpose and scope

R-089 reduced the controlled Cartan contribution to coefficient tails and proposed the global Sobolev ledger (3.12). It also described a rational same-root residual by appending a generic backward-heat and lower-chaos forest to a Taylor/Wick packet. This note stress-tests both steps before they are used in the full progressive assembly.

Four conclusions are proved.

1. The two Cartan coefficients of R-089 are not independent: $b_{A,j} = \nabla c_{A,j}$. Their Fourier multipliers recombine exactly to $iq\hat{c}(q-p)$. Thus the complete controlled Cartan trace is a single conservative coefficient trace.
2. The cutoff-uniform, unprojected global Sobolev ledger R-089 (3.12) is false for every $s > 0$ in the pinned sharp-cube/A1 setting. A fixed active control shell produces a current-root first Gaussian chaos whose contribution is bounded below once per root. The witness remains root-diagonal and is removed by the relative FAR projector, so it does not refute R-089 (3.9) or controlled Cartan CFAR.
3. The R-089 switch is centered and covariance matched only before conditioning, not conditionally on its branch event. More strongly, the complete local scalar raw rational endpoint has an exact negative adapted fixture. This rules out automatic algebraic positivity of the local endpoint, but not Cameron–Martin or sextic absorption and not the full torus packet.
4. R-063 lower chaoses reconstruct the literal coefficient–Wick product; they are not an additional physical term. The rational successor must begin from the complete endpoint and use the nonduplicating R-079 temporal decomposition. Three non-interchangeable proof obligations remain: projected Cartan FAR, complete signed NEAR, and progressive assembly inside OVERLAP.

The host claim stays T4. This note proves neither the projected coefficient tail, the complete signed rational/linear NEAR form, REG, progressive assembly, uniform OVERLAP, R-066 one-use in that assembly, Nelson, an interacting measure, nor Sector-A closure. It does not invalidate R-089’s progressive covariance compression, exact Fourier trace, quartic terminal lemma, or conditional Taylor algebra.

The load-bearing authorities are A1, R-063, R-066/R-070, R-078, R-079, R-084, R-086, and R-089. A1 supplies the two-sided order-four root symbol on the sharp cube. R-063 supplies exact Wick reconstruction. R-079 supplies the complete-current temporal identity. R-086 fixes the paid coefficient-dominant orientations. No open A11 theorem is used; only its already accepted finite-cutoff algebra is inherited.

2 Exact conservative Cartan compression

Use the R-089 notation

$$Z_j^- = w_j + A^{(\ell)}, \quad a^{[j]} = A^{(j)} - A^{(\ell)}, \quad Z_j^+ = Z_j^- + a^{[j]}, \quad (2.1)$$

and, for a production map $\Phi = \Phi_{A,j}$,

$$c_{A,j} = \Phi(Z_j^+) - \Phi(Z_j^-). \quad (2.2)$$

R-089 defined

$$\begin{aligned} b_{A,j,i} &= [D\Phi(Z_j^+) - D\Phi(Z_j^-)]^T \partial_i Z_j^- \\ &\quad + D\Phi(Z_j^+)^T \partial_i a^{[j]}. \end{aligned} \quad (2.3)$$

Theorem 2.1 (conservative coefficient identity)

At every finite cutoff and positive floor,

$$\boxed{b_{A,j,i} = \partial_i c_{A,j}, \quad L_{A,j} v = \nabla(c_{A,j}^T v).} \quad (2.4)$$

If $P_t^{(j)}$ is the current-root Ornstein–Uhlenbeck semigroup and $c_{A,j,t} = P_t^{(j)} c_{A,j}$, then

$$b_{A,j,t,i} = \partial_i c_{A,j,t}. \quad (2.5)$$

Proof. Differentiate (2.2). The two chain-rule terms are exactly (2.3), with no remainder. Substitution in the R-089 first-order operator gives the second identity. Spatial derivatives commute with the probability-space OU semigroup, proving (2.5). $\square$

For a root covariance eigenvector $v_{p,\mu} = \sqrt{\lambda_{j,p}} e^{ipx} e_\mu$ and output q , put $r = q - p$. Equation (2.5) gives

$$\widehat{b}_{A,j,t,i}(r) + ip_i \widehat{c}_{A,j,t}(r) = ir_i \widehat{c}_{A,j,t}(r) + ip_i \widehat{c}_{A,j,t}(r) = iq_i \widehat{c}_{A,j,t}(r). \quad (2.6)$$

Therefore R-089 (3.7) sharpens to the exact one-coefficient identity

$$\boxed{\|\Pi_m L_{A,j,t} \Gamma_j^{1/2}\|_{\text{HS}}^2 = \sum_{\mu,p} \lambda_{j,p} \sum_q |\pi_m(q)|^2 |q|^2 |\widehat{c}_{A,j,t}(q-p)^T e_\mu|^2.} \quad (2.7)$$

This equality is unconditional and keeps every physical-shell cross term.

Under the pinned LP convention

$$|\pi_m| \leq 1, \quad |q|_\infty \leq 2^{m+1}, \quad (2.8)$$

one has $|q|^2 \leq 12 \cdot 2^{2m}$. With $\Lambda_{0,j} \leq \kappa_0 2^{-j}$ and the R-084 prefactor $3/(40P)$,

$$\boxed{S_C^{\text{ctrl}} \leq \frac{9\kappa_0}{10P} \sum_{A,j} \int_0^\infty e^{-2t} 2^{-j} \sum_{m \geq j+C} 2^{2m} \mathbb{E} \|\widetilde{\Pi}_m c_{A,j,t}\|_2^2 dt.} \quad (2.9)$$

The constant 9/10 is conditional on (2.8); the exact q -weighted trace (2.7) is not. Formula (2.9) is the corrected one-coefficient localized CFAR target.

3 The global unprojected Sobolev ledger fails

By (2.4), the R-089 quantity (3.11) is exactly

$$\begin{aligned} \mathfrak{E}_s(J) &= \sum_{A,j \leq J} \int_0^\infty e^{-2t} \mathbb{E} [\kappa_0 2^{-(2s+1)j} \|c_{A,j,t}\|_{H^{s+1}}^2 \\ &\quad + \kappa_1 2^{(1-2s)j} \|c_{A,j,t}\|_{H^s}^2] dt. \end{aligned} \quad (3.1)$$

Theorem 3.1 (uniform no-go for every positive exponent)

For every $s > 0$, there is a fixed finite-energy cylindrical-simple deterministic control and constants $c_s > 0$, $j_* > 0$ such that

$$\boxed{\mathfrak{E}_s(J) \geq c_s(J - j_*) \quad (J \geq j_*)}. \quad (3.2)$$

For the same fixture,

$$\sup_J \{1 + \mathbb{E}[X_J^{1/2} Y_J^{1/2}]\} < \infty. \quad (3.3)$$

Consequently no $s > 0$ makes the cutoff-uniform claim R-089 (3.12) true for the full stated control class.

Proof. First isolate a genuinely nonlinear production direction. On a normalized eigenray,

$$F_e(x) = \frac{x^3}{e + x^2}, \quad F_e''(x) = -\frac{2ex(x^2 - 3e)}{(e + x^2)^3}, \quad F_1''(1) = \frac{1}{2}. \quad (3.4)$$

This is only an unsmoothed nonaffinity witness. Let P_{C_∞} denote Gaussian convolution by all other roots and the accepted target heat. Gaussian convolution is injective because its Fourier multiplier never vanishes. Hence $P_{C_\infty} F_A$ is not affine. Choose a finite-low background z_0 and directions u, h so that

$$M_\infty = D^2(P_{C_\infty} F_A)(z_0)[u, h] \neq 0. \quad (3.5)$$

Finite-cutoff covariance convergence gives $|M_j| \geq m_* > 0$ for all large j . No claim is made that the particular point in (3.4) survives heat; (3.5) is the load-bearing generic choice.

Next choose a fixed deterministic control in one active shell $k_0 \geq j_0$, not solely in the subtracted finite-low bucket:

$$a^{[j]}(x) = \delta a_0(x)u \quad (j \geq k_0), \quad (3.6)$$

where a_0 is a real trigonometric mode and $\delta > 0$ is fixed and small. The finite-low background producing z_0 is kept separately. Thus the same active secant persists once in every later root coefficient while the control CM and spatial costs do not grow with J .

Whiten the current root g_j^h . Project $c_{A,j}$ onto the first Wiener chaos of this root and onto constants in every other root. Gaussian integration by parts and a Taylor expansion in δ give

$$\text{Proj}_{g_j^h}^{(1)} c_{A,j} = \delta M_j a_0 g_j^h + O(\delta^2) \quad (3.7)$$

uniformly for large j . Choose δ so the error is at most half of the main coefficient. The current-root semigroup multiplies (3.7) by e^{-t} ; this is distinct from the injective target heat used in (3.5).

The A1 two-sided order-four symbol and the sharp-cube convention provide an interior annular subset S_j° of positive density for which

$$\sum_{p \in S_j^\circ} \lambda_{j,p} \geq \underline{\kappa}, 2^{-j}. \quad (3.8)$$

For a real field, retain one member of each antipodal pair and its cosine or sine coordinate; fixed multiplication by a_0 causes only finitely many collisions, and deleting them preserves positive density. Orthogonal chaos projection and (3.7)–(3.8) then imply

$$\mathbb{E} \|c_{A,j,t}\|_{H^s}^2 \geq K_s \delta^2 e^{-2t} 2^{(2s-1)j}. \quad (3.9)$$

Since $b = \nabla c$, the other channel obeys the corresponding lower bound $K_{s+1} \delta^2 e^{-2t} 2^{(2s+1)j}$.

Both weights in (3.1) cancel the shell powers:

$$2^{(1-2s)j} 2^{(2s-1)j} = 1, \quad 2^{-(2s+1)j} 2^{(2s+1)j} = 1. \quad (3.10)$$

Moreover $\int_0^\infty e^{-2t} e^{-2t} dt = 1/4$. Summation over large roots proves (3.2). Finally the fixed smooth control has cutoff-independent CM cost; the A1 order-four Gaussian shell exponent is geometrically summable,

giving uniform point moments and hence a uniform terminal L^6 moment. This proves (3.3) and the contradiction. $\square$

The theorem fires [NG-2026-07-25-A13-GLOBAL-UNPROJECTED-CARTAN-COEFFICIENT-LEDGER](#). Its scope is exact: the bad coefficient frequencies lie at $j + O(1)$. For a sufficiently large fixed gap, $Q_{j,C-2}^{\text{coef}}$ removes this selected component. The numerical FFT fixture verifies only that component's near-root support; it does not assert that the complete nonlinear coefficient has zero far tail. Thus (3.2) falsifies the unprojected relaxation (3.12), not the projected estimate (3.9).

The correct successor is the expectation-inside localized estimate

$$\sum_{A,j} \int_0^\infty e^{-2t} 2^{-j} \sum_{m \geq j+C} 2^{2m} \mathbb{E} \|\tilde{\Pi}_m P_t^{(j)} c_{A,j}\|_2^2 dt \lesssim 2^{-2\beta C} \{1 + \mathbb{E}[X^{1/2} Y^{1/2}]\} \quad (3.11)$$

for some $\beta > 0$, with the relative projector retained until after the root sum. An elementary excess-tail extraction beginning five coefficient shells above the root is available, but the uniform bound on that excess ledger is still open.

4 Rational conditioning correction and raw endpoint sign

Let $G \sim N(0, 1)$, $Q = G^2 - 1$, and $E_+ = \{|G| \geq 1\}$ with probabilities $p_+ = 2\bar{\Phi}(1)$ and $p_- = 1 - p_+$. For $\mathcal{H} = \sigma(E_+)$, symmetry gives conditional mean zero, but

$$V_+ = 1 + \frac{2\varphi(1)}{p_+} = 2.525135276160981\dots, \quad V_- = 1 - \frac{2\varphi(1)}{p_-} = 0.291125094772793\dots \quad (4.1)$$

Only $p_+ V_+ + p_- V_- = 1$. Therefore R-089 (5.9) is unconditionally centered and covariance matched; it does not satisfy the conditional $V = \Gamma$ hypothesis of R-089 Theorem 5.1. This attribution correction is included in [AUDIT-2026-07-25-A13-R089-RATIONAL-FOREST-DISJOINTNESS](#).

There is nevertheless a stronger local endpoint statement that needs no Taylor attribution. Put

$$b_e(u) = 4c_1 \left(u - \frac{5}{9} \frac{u^3}{u^2 + e} \right)^2, \quad z = \sqrt{e}, \quad a = \sqrt{e} C(G), \quad (4.2)$$

where $C = -1/2$ on $Q \geq 0$ and $C = +1/2$ on $Q < 0$, and take $Da = 0$ in this local scalar fixture. Direct arithmetic gives

$$\frac{b_e(z)}{4c_1 e} = \frac{169}{324}, \quad \frac{b_e(z - \sqrt{e}/2)}{4c_1 e} = \frac{16}{81}, \quad \frac{b_e(z + \sqrt{e}/2)}{4c_1 e} = \frac{144}{169}. \quad (4.3)$$

Hence

$$\Delta W_R = \frac{1}{2} [b_e(z + a) - b_e(z)] Q < 0 \quad \text{for } Q \neq 0, \quad (4.4)$$

and, using $\mathbb{E}[Q \mathbf{1}_{Q \geq 0}] = 2\varphi(1)$,

$$\boxed{\mathbb{E} \Delta W_R = -\frac{35840}{13689} c_1 e \varphi(1) = -0.633518209275045\dots, c_1 e.} \quad (4.5)$$

The symbolic c_1 dependence in (4.5) is essential. A decimal multiple of e alone is valid only after separately pinning c_1 .

For the smooth bounded even rule

$$C_M(G) = -\frac{1}{2} \tanh(M(G^2 - 1)), \quad (4.6)$$

write $h(s) = s(9 + 4s^2)/[9(1 + s^2)]$. Since

$$h'(s) = \frac{9 + 3s^2 + 4s^4}{9(1 + s^2)^2} > 0 \quad (s > 0), \quad (4.7)$$

the integrand in (4.4) stays negative for every $M > 0$. Dominated convergence recovers (4.5). The independent executable checks $M = 2, 4, 8$ numerically; the universal statement follows from (4.7), not from those samples.

This is a complete local scalar raw endpoint, containing its current square, Wick trace, and exact R-063 reconstruction once. It is not a torus/A1 strict-past control counterexample. It omits realization as $T_J v$, global $c = Da$, spatial derivative channels, six-component frame, regulator, paid subtractions, low objects, and exact CM/sextic costs. Those positive costs can absorb a fixed local loss. Small future covariance preserves the local negative expectation by continuity, but no production covariance threshold is claimed.

5 Forest nonduplication and canonical rational target

R-063's lower chaoses are an identity reconstructing a literal Wick product, not extra energy. The scalar diagnostic

$$G^2(G^2 - 1) = H_4(G) + 5H_2(G) + 2 \quad (5.1)$$

makes the rule visible: if the left side is already present, adding $5H_2 + 2$ again changes the endpoint. Consequently R-089's undefined $\mathcal{F}$ is unsafe. If $L : Q$ denotes the complete R-063 reconstructed product, no R-063 forest is appended. If it denotes a declared top-chaos or Bony piece, its complement must be defined exactly by subtraction from the complete endpoint. This is a bookkeeping audit, not a claim that every possible interpretation of $\mathcal{F}$ double-counts.

The canonical rational row is

$$C_R(z) = 2\sqrt{c_1} g(z)^T, \quad B_R(z) = C_R(z)^T C_R(z), \quad (5.2)$$

with current and heat-transported endpoint

$$J_R(H)_i = C_R(U + H)(D_i U + D_i H), \quad (5.3)$$

$$W_R(H) = \frac{1}{2} \|J_R(H)\|^2 - \frac{1}{2} \sum_i \int B_R(U + H) : \Gamma_{J,i}. \quad (5.4)$$

For the R-079 fields $A^*, A^0, A^{(j)}$ and $d_j = P_j - P_{j-1}$, define

$$f_j^R = d_j [J_R(A^{(j)}) - J_R(A^0)], \quad i_j^R = d_j [J_R(A^*) - J_R(A^{(j)})]. \quad (5.5)$$

The rational projection of the R-079 identity is

$$\boxed{\mathbb{E}[W_R(A^*) - W_R(A^0)] = L_\ell^R + \sum_j (F_j^R + I_j^R)}, \quad (5.6)$$

where

$$\begin{aligned} I_j^R &= \mathbb{E} \left[\langle d_j J_R(A^{(j)}), i_j^R \rangle + \frac{1}{2} \|i_j^R\|^2 \right] \\ &\quad - \frac{1}{2} \mathbb{E} \sum_i \int [B_R(A^*) - B_R(A^{(j)})] : \Delta \Gamma_{j,i}. \end{aligned} \quad (5.7)$$

The current difference contains both coupled channels

$$J_R(A^*) - J_R(A^{(j)}) = [C_R(U + A^*) - C_R(U + A^{(j)})](DU + DA^{(j)}) + C_R(U + A^*)D(A^* - A^{(j)}). \quad (5.8)$$

Thus coefficient Jensen defect, derivative feedback, square, and trace may not be estimated as unrelated additions.

After the canonical R-086 paid subtraction, the remaining rational target is

$$\boxed{L_\ell^R + \sum_j (F_j^R + I_j^R) - \mathbb{E}[R_{\leq, R}(A^*) - R_{\leq, R}(A^0)]}. \quad (5.9)$$

The missing identification is to prove that the R-086 coefficient-dominant Bony projection commutes with this R-079 decomposition without duplication. R-066/R-070 supply the already heat-transported endpoint, while R-079 decomposes it temporally; they are not independent terms to append to one another.

6 Complete packet and the remaining proof architecture

At the full endpoint, the complete unpaid packet is the exact identity

$$\begin{aligned} \mathcal{U}_{J, \pi} &:= \mathbb{E}B_{\text{alg}}^>(A^0) + L_\ell + \sum_k (F_k + I_k) \\ &\quad - \mathbb{E}[R_{\leq}(A^*) - R_{\leq}(A^0)] = \mathbb{E}B_{\text{alg}}^>(A^*). \end{aligned} \quad (6.1)$$

The R-080 revisit fixture shows why no isolated low-object inequality is available: its separate low loss is of order $-ct^4/2 + O(t^3)$ while the displayed costs and trace are $O(t^2)$, and cancellation can occur only in the complete packet.

A sufficient modular route now has three non-interchangeable proof obligations.

H_C (projected Cartan FAR). Prove (3.11), or the exact q -weighted version (2.7), with $Q_{j, C-2}^{\text{coef}}$ retained and a cutoff-uniform far-gap gain.

H_N (complete signed NEAR). Bound the full reconstructed rational and linear Pauli–Fierz NEAR packet, with the literal Wick endpoint present once and all endpoint-square, native heat, temporal, paid, and low companions explicitly mapped.

H_A (progressive assembly). Recombine the complete packet uniformly in cutoff, partition, and revisit, preserving the R-079 low and future-feedback cross terms and every one-use budget inside full OVERLAP.

These are not asserted to be a pairwise-logically-independent theorem. H_C does not control the same-root rational sign defect. A componentwise H_N does not presently imply H_C . Conversely, cross-cancellation in a future full endpoint proof could close H_A without proving the two pieces separately. What is proved is the non-implication needed for workflow: closing one named component does not by itself establish complete assembly.

The remaining arithmetic is compatible but not a proof. With the already reserved charges,

$$\frac{27}{100} - \frac{6}{100} - \frac{15}{100} = \frac{3}{50}, \quad \frac{1}{2(11/10)} - \frac{9}{20} = \frac{1}{220}, \quad \frac{10}{9} - \frac{11}{10} = \frac{1}{90}. \quad (6.2)$$

Thus the sextic, control, and exponent ledgers have positive formal margins. No estimate in H_C , H_N , or H_A is inferred from those margins.

7 Attempt and evidence map

1. **Two Cartan coefficient tails – superseded.** The chain rule proves $b = \nabla c$ and the exact conservative trace (2.7). This is a successful simplification.
2. **Global Sobolev extraction – falsified.** The fixed active-shell/current-root first-chaos witness makes every positive s critical and gives linear root accumulation.
3. **Past-root witness – unnecessary branch.** A past-root chaos would have OU time integral $1/2$; the registered proof uses the current root, whose e^{-t} attenuation gives $1/4$ and matches the executable.
4. **Fixed unsmoothed Hessian – insufficient.** $F_1''(1) = 1/2$ detects nonaffinity but does not survive target heat by fiat. Injectivity of Gaussian convolution supplies the generic smoothed background and directions used in (3.5).
5. **Projected coefficient tail – viable and open.** The no-go witness is root-diagonal and the FAR projection deletes it. No counterexample to (2.9) or (3.11) was found.
6. **Conditional covariance match – corrected.** The two branch variances are (4.1), although their weighted average is one.
7. **Taylor-fragment positivity – failed.** R-089 already found a negative Taylor direction; (4.5) strengthens the local statement to the unexpanded raw scalar endpoint.
8. **Automatic forest repair – falsified as a method.** R-063 reconstructs rather than supplements the product. A generic extra forest is not a well-defined cancellation mechanism.
9. **Canonical temporal recombination – identified.** Equations (5.2)–(5.9) give the nonduplicating rational successor; its bound is not proved.
10. **REG/OVERLAP shortcut – rejected.** Projected FAR and signed NEAR do not by themselves pay low/revisit assembly. The sufficient roadmap is H_C , H_N , then H_A inside OVERLAP.

8 Devil’s-advocate review

1. **Objection:** $b = \nabla c$ holds only before OU averaging. **DISMISSED.** Spatial differentiation commutes with the root OU semigroup, so (2.5) is exact.
2. **Objection:** the $9/10$ prefactor is universal. **VALID WITH MITIGATION.** It uses the pinned upper LP support (2.8). The exact trace (2.7) remains valid without that convention.
3. **Objection:** the control witness lies only in $A^{(\ell)}$ and cancels from the secant. **DISMISSED.** The secant direction occupies a fixed active shell $k_0 \geq j_0$ and persists in $a^{[j]}$ for all later roots. The finite-low field is a separate background.
4. **Objection:** $F_1''(1) = 1/2$ proves the smoothed chaos coefficient is nonzero at the same point. **UPHELD AND REPAIRED.** The proof does not make that inference; Gaussian-heat injectivity gives some generic z_0, u, h , followed by finite-cutoff convergence.
5. **Objection:** the time factor should be $1/2$. **DISMISSED FOR THE REGISTERED FIXTURE.** The witness is current-root first chaos, so the squared OU attenuation supplies another e^{-2t} and the integral is $1/4$. A past-root construction would indeed use $1/2$.

6. **Objection:** real Fourier antipodes can cancel the lower bound. **DISMISSED.** Select one cosine/sine representative from each pair and delete the finite collision set generated by the fixed multiplier. A positive-density interior subset remains.
7. **Objection:** Theorem 3.1 disproves controlled Cartan CFAR. **DISMISSED.** It disproves only the stronger unprojected Sobolev relaxation. Its selected component is annihilated by the relative FAR projector; the complete projected tail stays open.
8. **Objection:** the switch satisfies the conditional theorem because $\mathbb{E}G^2 = 1$. **DISMISSED.** Conditional variances are (4.1). With the trivial sigma-field the coefficient switch is not measurable.
9. **Objection:** the raw endpoint is already a full A13 counterexample. **DISMISSED.** It is a local scalar sign fixture and omits the global control realization, spatial packet, paid terms, CM, sextic, and low objects.
10. **Objection:** R-063 lower chaoses may always be appended as a positive forest. **DISMISSED.** They reconstruct the product. Adding them to an unexpanded endpoint gives the explicit double count (5.1).
11. **Objection:** $H_C + H_N$ establishes progressive OVERLAP. **DISMISSED.** R-079 future feedback and R-080 low/revisit cancellation remain in H_A . The modular route is sufficient architecture, not an iff.

9 Executable verification and no-overclaim statement

The primary executable checks the conservative Fourier identity, LP constant, exact shell-power cancellation, current-root OU integral, rational fractions, conditional variances, Hermite reconstruction, R-079 polarization, and budget arithmetic. The independent executable shares no imports with it; it uses explicit shell loops, random complex Fourier frequencies, high-precision Gaussian integration, smooth endpoint quadrature, Gauss–Hermite projection, and separate packet polarization. Run

```
python codes/foundations/a13_classii_cartan_excess_tail_rational_forest_nonduplication_boundary_verify.py
```

R-090 proves the conservative single-coefficient trace, falsifies only the global unprojected cutoff-uniform ledger R-089 (3.12), corrects the conditional covariance attribution, proves a local raw rational endpoint sign no-go, and repairs forest/temporal bookkeeping. Projected CFAR, complete signed NEAR, REG, progressive assembly/full OVERLAP, one-use in that assembly, Nelson, measure construction, and Sector A remain open. Tier stays T4; no promotion is made.

Result footer

Result ID: A13-CLASSII-GLOBAL-UNPROJECTED-CARTAN-COEFFICIENT-
LEDGER-NOGO-RATIONAL-FOREST-BOUNDARY

Ledger: R-090.

Precise statement: The complete controlled Cartan trace compresses exactly to one conservative coefficient. The global unprojected Sobolev ledger proposed in R-089 (3.12) fails for every $s > 0$ by a fixed active-shell/current-root first-chaos witness, while the relative FAR target survives. The R-089 branch is not conditionally covariance matched; the local raw rational endpoint is strictly negative. R-063 forests reconstruct once and R-079 is the canonical nonduplicating temporal assembly.

Scope: Finite cutoff, positive floor, pinned sharp-cube/A1

conventions. The no-go concerns the unprojected ledger, not projected CFAR or the full A13 form.

Dependencies: A1; R-063; R-066/R-070; R-078; R-079; R-084; R-086; R-089.

Evidence grade: ANALYTIC, EXACT, EXECUTED, BOUNDARY.

Reproduction command: python codes/foundations/a13_classii_cartan_excess_tail_rational_forest_nonduplication_boundary_verify.py

Expected output: Primary 57/57; independent 52/52; integrated 152/152; aggregate 261/261 PASS; exit zero.

Audit: AUDIT-2026-07-25-A13-R089-RATIONAL-FOREST-DISJOINTNESS.

Negative result: NG-2026-07-25-A13-GLOBAL-UNPROJECTED-CARTAN-COEFFICIENT-LEDGER.

Falsification gate: Failure of chain-rule/Fourier compression, sharp-cube/A1 shell scaling, chaos accumulation, rational arithmetic, forest reconstruction, packet polarization, evidence hashes, PDF, or counts.

Tier before / after: T4 / T4; no promotion.

No-overclaim statement: No projected Cartan CFAR, complete signed NEAR, REG, progressive assembly, full OVERLAP, one-use, Nelson, interacting measure, removal limit, or Sector-A closure.

Next required action: Prove H_C with relative projection retained; define and prove the nonduplicating complete signed H_N; then prove H_A inside full OVERLAP.